Status	Finished
Started	Sunday, 11 October 2026, 4:43 PM
Completed	Sunday, 11 October 2026, 4:49 PM
Duration	5 mins 59 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99

b. 90 and 99 ✓

c. 90 and 94

d. 89 and 94

Question **2**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search

b. Binary search

c. Linear search ✓

d. Interpolation search

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓

b. There must be a mechanism to access middle element directly

c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

d. Must use a sorted array

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O👎
c. O(log n) ✓
d. O(n log n)

Question **5**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The list is sorted
c. The search ends successfully ✓
d. The search continues in the right sublist

Question **6**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Binary ✓
b. Both (1) & (3)
c. Linear
d. Hash

Question **7**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. When the item is somewhere in the middle of the array ✓

c. When the item is the last element in the array

d. Item is the last element in the array or item is not there at all

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓

b. The list must be sorted

c. There should be direct access to the middle element in any sublist

d. Number values should only be present

Number values should only be present

Question **9**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search stops immediately ✓

b. The search continues until the end of the list

c. The search starts over from the beginning

d. The search backtracks to find duplicate elements

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Skipping every second element
- c. Sorting the list before searching
- d. Dividing the list in half repeatedly

Question **11**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The higher middle element is chosen as the middle element
- b. The lower middle element is chosen as the middle element ✗
- c. The search stops
- d. The middle element is chosen randomly

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The list is sorted
- b. The search stops
- c. The search continues from the beginning
- d. The search continues in the left or right sublist ✓

Question **13**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓

b. Complexity

c. Selection

d. Insertion

Question **14**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1

b. log n

c. n ✓

d. n/2

Question **15**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element

b. Binary search has a lower time complexity on large, sorted lists ✓

c. Binary search works on unsorted lists

d. Binary search does not require dividing the list